

ROLL & DEAL · ORIGINAL TABLE GAMES

# Game sheets

*Dice rolled by the player. Cards dealt by the dealer.*

---

**Dice Spread**

RTP 96.30% · Between

---

**Moving Target**

RTP 96.15% · Exact Hit

---

**Mirror**

RTP 94.91% · Mirror

---

**Lock & Roll**

RTP 95.93% · Lock & Roll

---

One sheet per game: rules, examples, the paytable and the mathematics. The paytables and declared figures are generated from each game's code and proven by exact enumeration and seeded Monte Carlo runs. RTPs shown here are those of each main bet; the sheets give every bet's. Build c07029bd19a3c471.

# Dice Spread

Will the card land between your dice? · original table game · **status: playable**

## Summary

The player places bets and rolls two dice; the dealer deals one card from a shoe of aces to sixes. The main bet, **Between**, wins when the card falls strictly between the two dice, and pays more the closer together they are. Four optional side bets settle on the same roll and card. There are no decisions after the bets.

## Rules

### Equipment

- Two six-sided dice, rolled by the player.
- A shoe of six decks, each with the ace to six in four suits: 144 cards. The ace counts as 1 and suits do not matter.
- The cut card sits with a quarter of the shoe left. When it comes out, the round finishes and the shoe is reshuffled before the next round.

### A round

1. **Bet.** Place Between and, if you like, any of the side bets. Side bets need a Between bet. Limits: Between 0.50 to 250.00; each side bet 0.50 to 25.00.
2. **Roll.** Roll both dice. The *spread* is the high die minus the low die: 0 for a pair, 5 for a one and a six.
3. **Card.** The dealer deals one card face down, then turns it over.
4. **Settle.** Every bet settles on that roll and card.

### Between

Between wins when the card is strictly between the two dice. The payout depends on the spread:

| Spread       | Cards that win | Pays   |
|--------------|----------------|--------|
| 2            | 1 value        | 4 to 1 |
| 3            | 2 values       | 2 to 1 |
| 4            | 3 values       | 1 to 1 |
| 5            | 4 values       | 1 to 2 |
| 1, or a pair | none           | push   |

A card equal to either die, or outside the dice, loses. On a push the stake is returned.

### Side bets

| Bet      | Wins when                                                | Pays    |
|----------|----------------------------------------------------------|---------|
| Match    | the card equals either die                               | 2 to 1  |
| Bullseye | the spread is exactly 2 and the card is the middle value | 22 to 1 |
| Doubles  | the dice are a pair                                      | 4 to 1  |
| Triple   | the dice are a pair and the card matches it              | 30 to 1 |

### Examples

- Dice 2 and 5 (spread 3), card 3: Between wins 2 to 1; Match loses.

- Dice 4 and 6 (spread 2), card 5: Between wins 4 to 1 and Bullseye wins 22 to 1.
- Dice 3 and 3, card 3: Between pushes; Match, Doubles and Triple win 2, 4 and 30 to 1.
- Dice 1 and 6 (spread 5), card 6: Between loses; Match wins 2 to 1.

## Bets and payable

| Bet             | RTP    | House edge | Hit frequency | Push   | Max exposure | Volatility index | Limits        |
|-----------------|--------|------------|---------------|--------|--------------|------------------|---------------|
| Between (main)  | 96.30% | 3.70%      | 18.52%        | 44.44% | 4×           | 1.117            | 0.50 – 250.00 |
| Match (side)    | 91.67% | 8.33%      | 30.56%        | —      | 2×           | 1.382            | 0.50 – 25.00  |
| Bullseye (side) | 85.19% | 14.81%     | 3.70%         | —      | 22×          | 4.344            | 0.50 – 25.00  |
| Doubles (side)  | 83.33% | 16.67%     | 16.67%        | —      | 4×           | 1.863            | 0.50 – 25.00  |
| Triple (side)   | 86.11% | 13.89%     | 2.78%         | —      | 30×          | 5.094            | 0.50 – 25.00  |

RTP and house edge are per unit staked, pushes included. Hit frequency is the chance that a bet wins in a round. Push is the chance that the stake is simply returned. Max exposure is the largest net win per unit staked. The volatility index is the standard deviation of the net result per unit staked.

### Between (main bet)

Wins when the card falls strictly between the two dice, at odds set by the spread (high die minus low die). A pair or dice one apart push.

| Outcome                | Pays   | Probability |
|------------------------|--------|-------------|
| Spread 2, card between | 4 to 1 | 3.704%      |
| Spread 3, card between | 2 to 1 | 5.556%      |
| Spread 4, card between | 1 to 1 | 5.556%      |
| Spread 5, card between | 1 to 2 | 3.704%      |
| Spread 1, or a pair    | Push   | 44.444%     |

### Match (side bet)

Wins when the card equals either die.

| Outcome            | Pays   | Probability |
|--------------------|--------|-------------|
| Card matches a die | 2 to 1 | 30.556%     |

### Bullseye (side bet)

Bull's eye: wins when the dice are two apart and the card is the value between.

| Outcome                            | Pays    | Probability |
|------------------------------------|---------|-------------|
| Spread 2, card is the middle value | 22 to 1 | 3.704%      |

### Doubles (side bet)

Wins when the dice show a pair.

| Outcome             | Pays   | Probability |
|---------------------|--------|-------------|
| The dice are a pair | 4 to 1 | 16.667%     |

### Triple (side bet)

Wins when the dice show a pair and the card matches it.

| Outcome                         | Pays    | Probability |
|---------------------------------|---------|-------------|
| A pair, and the card matches it | 30 to 1 | 2.778%      |

## Mathematics

The tables above are generated from the game's code, which the tests below verify. Method and conventions: `docs/MATH.md`.

### Card source

The declared figures treat the card as uniform over ace to six, as an infinite shoe would deal it. The six-deck shoe returns exactly the same in the long run to a player who bets the same way every round. Each round deals one card and the cut card always comes out after 108 rounds, so the rounds a shoe deals never depend on the cards. Over all shuffles, the card of any given round is therefore equally likely to be each value.

### Exact test

`dice-spread.test.ts` runs the game over all 36 rolls  $\times$  6 card values (216 outcomes). Every figure must equal the enumerated fraction:

| Bet      | RTP   | House edge | Hit frequency  | Variance $\sigma^2$ | Volatility index $\sigma$ |
|----------|-------|------------|----------------|---------------------|---------------------------|
| Between  | 26/27 | 1/27       | 5/27, push 4/9 | 3641/2916           | 1.117                     |
| Match    | 11/12 | 1/12       | 11/36          | 275/144             | 1.382                     |
| Bullseye | 23/27 | 4/27       | 1/27           | 13754/729           | 4.344                     |
| Doubles  | 5/6   | 1/6        | 1/6            | 125/36              | 1.863                     |
| Triple   | 31/36 | 5/36       | 1/36           | 33635/1296          | 5.094                     |

The volatility index is the standard deviation of the net result per unit staked. The test repeats the enumeration with an infinite shoe, suits included ( $36 \times 24 = 864$  draws), and gets the same fractions.

All of Between's house edge comes from its spread-2 line. With one winning value in six, true odds are 5 to 1 and the line pays 4 to 1. The spread 3, 4 and 5 lines pay exactly true odds.

### Monte Carlo test

`dice-spread.math.test.ts` (run with `pnpm test:math`) plays 124,852,059 rounds of the full layout, all five bets at 0.50, dealt from the real six-deck shoe with the seeded generator (seed `dice-spread/monte-carlo`). That count makes  $\pm 0.15$  percentage points equal to 3.29 standard errors for the most volatile bet, Triple. Every RTP must land within  $\pm 0.15$  pp of its declared value, and every hit and push frequency within 3.29 binomial standard errors.

| Bet      | Simulated RTP | Declared RTP | Difference | Standard error |
|----------|---------------|--------------|------------|----------------|
| Between  | 96.289%       | 96.296%      | -0.007 pp  | 0.010 pp       |
| Match    | 91.676%       | 91.667%      | +0.009 pp  | 0.012 pp       |
| Bullseye | 85.205%       | 85.185%      | +0.020 pp  | 0.039 pp       |
| Doubles  | 83.376%       | 83.333%      | +0.043 pp  | 0.017 pp       |
| Triple   | 86.125%       | 86.111%      | +0.014 pp  | 0.046 pp       |

### Card counting

A player who bets the same way every round cannot beat the shoe, but one who tracks the cards dealt can choose when to bet big. Between's value depends on what is left in the shoe: an ace or a six can never fall between the dice, while a three or a four often does. Over the 36 rolls, one unit on Between is

worth  $-5/9$  when the card is an ace or a six,  $+1/12$  for a two or a five, and  $+13/36$  for a three or a four. Once many aces and sixes have gone, the rest of the shoe can favour the player.

`counting.test.ts` computes the exposure exactly, for a player who knows the composition of the shoe before every round:

| Penetration   | Rounds per shoe | Rounds favouring the player | Player edge in them | Break-even bet spread |
|---------------|-----------------|-----------------------------|---------------------|-----------------------|
| 75% (default) | 108             | 8.57%                       | +2.05%              | 22 : 1                |
| 50%           | 72              | 3.66%                       | +1.18%              | 87 : 1                |
| 25%           | 36              | 0.47%                       | +0.59%              | over 1,000 : 1        |

The table limits (0.50 to 250.00) allow a 500 : 1 spread, so at the default penetration a skilled counter could beat Between wherever the cards dealt are visible, as at a live table. A live table should shuffle after every round, use a continuous shuffler, or cut the penetration well below 50%. An RNG table that reshuffles every round returns exactly the declared figures.

Bullseye also depends on the composition, but favours the player in only 0.47% of rounds at the default penetration. Match, Doubles and Triple cannot be counted: every card value is worth the same to them.

## Change log

---

- 2026-09-30: first playable version, with rules, declared math, exact and Monte Carlo tests, a counting analysis and the table.
- 2026-09-30: renamed from its working title, with English names for the game and its bets.

# Moving Target

*Your dice set the target. Will the cards land on it?* · original table game · **status: playable**

## Summary

The player places bets and rolls two dice: their sum is the round's target. The dealer then deals cards from a shoe of aces to tens, one at a time and face up, adding them up, and stops as soon as the total reaches the target. The main bet, **Exact Hit**, wins when the total lands exactly on the target, and pays more on the targets that are harder to hit. Two optional side bets settle on the same deal. There are no decisions after the bets.

## Rules

### Equipment

- Two six-sided dice, rolled by the player.
- A shoe of six decks, each with the ace to the ten in four suits (40 cards, no picture cards): 240 cards. The ace counts 1 and every other card its number; suits do not matter.
- The cut card sits with a quarter of the shoe left. When it comes out, the round finishes and the shoe is reshuffled before the next round. A round deals at most 12 cards, so it never runs out.

### A round

1. **Bet.** Place Exact Hit and, if you like, the side bets. Side bets need an Exact Hit bet. Limits: Exact Hit 0.50 to 250.00; each side bet 0.50 to 25.00.
2. **Roll.** Roll both dice. Their sum, from 2 to 12, is the target.
3. **Deal.** The dealer deals cards face up, one at a time, adding them up, and stops as soon as the total reaches or passes the target.
4. **Settle.** Every bet settles on that target and those cards.

### Exact Hit

Exact Hit wins when the dealer's total lands exactly on the target; a total that passes it loses. It pays by target:

| Target | Pays     | Target | Pays     |
|--------|----------|--------|----------|
| 2      | 7.5 to 1 | 8      | 4 to 1   |
| 3      | 7 to 1   | 9      | 3.5 to 1 |
| 4      | 6 to 1   | 10     | 3 to 1   |
| 5      | 5.5 to 1 | 11     | 5 to 1   |
| 6      | 5 to 1   | 12     | 5 to 1   |
| 7      | 4.5 to 1 |        |          |

The generated tables below write the half-unit payouts as fractions: 15 to 2 is 7.5 to 1.

### Side bets

| Bet        | Wins when                                                   | Pays   |
|------------|-------------------------------------------------------------|--------|
| First Card | the first card alone is the target (impossible on 11 or 12) | 9 to 1 |
| 3+ Cards   | the dealer needs three cards or more                        | 4 to 1 |

## Examples

- Dice 3 and 4 (target 7), cards 3 and 4: the total lands on 7. Exact Hit wins 4.5 to 1; 3+ Cards loses, as two cards were enough.
- Dice 1 and 4 (target 5), card 5: Exact Hit wins 5.5 to 1 and First Card 9 to 1.
- Dice 6 and 6 (target 12), cards 10 and 5: the total, 15, passes the target, and every bet loses.
- Dice 5 and 6 (target 11), cards A, 2, 3 and 7: the total, 13, passes the target. Exact Hit loses; 3+ Cards wins.

## Bets and payable

| Bet               | RTP    | House edge | Hit frequency | Max exposure | Volatility index | Limits        |
|-------------------|--------|------------|---------------|--------------|------------------|---------------|
| Exact Hit (main)  | 96.15% | 3.85%      | 17.30%        | 7.5×         | 2.148            | 0.50 – 250.00 |
| First Card (side) | 91.67% | 8.33%      | 9.17%         | 9×           | 2.886            | 0.50 – 25.00  |
| 3+ Cards (side)   | 89.58% | 10.42%     | 17.92%        | 4×           | 1.917            | 0.50 – 25.00  |

RTP and house edge are per unit staked, pushes included. Hit frequency is the chance that a bet wins in a round. Max exposure is the largest net win per unit staked. The volatility index is the standard deviation of the net result per unit staked.

### Exact Hit (main bet)

Wins when the dealer's total lands exactly on the target (the sum of the dice), at odds set by the target.

| Target | Chance | Pays    | Hit frequency | House edge |
|--------|--------|---------|---------------|------------|
| 2      | 2.78%  | 15 to 2 | 11.00%        | 6.50%      |
| 3      | 5.56%  | 7 to 1  | 12.10%        | 3.20%      |
| 4      | 8.33%  | 6 to 1  | 13.31%        | 6.83%      |
| 5      | 11.11% | 11 to 2 | 14.64%        | 4.83%      |
| 6      | 13.89% | 5 to 1  | 16.11%        | 3.37%      |
| 7      | 16.67% | 9 to 2  | 17.72%        | 2.56%      |
| 8      | 13.89% | 4 to 1  | 19.49%        | 2.56%      |
| 9      | 11.11% | 7 to 2  | 21.44%        | 3.54%      |
| 10     | 8.33%  | 3 to 1  | 23.58%        | 5.68%      |
| 11     | 5.56%  | 5 to 1  | 15.94%        | 4.38%      |
| 12     | 2.78%  | 5 to 1  | 16.53%        | 0.81%      |

Chance is the share of rounds with each target; the hit frequency and house edge are for the rounds with that target.

### First Card (side bet)

Wins when the first card alone is the target. A target of 11 or 12 cannot win.

| Outcome                      | Pays   | Probability |
|------------------------------|--------|-------------|
| The first card is the target | 9 to 1 | 9.167%      |

### 3+ Cards (side bet)

Wins when the dealer needs three cards or more to reach the target.

| Outcome             | Pays   | Probability |
|---------------------|--------|-------------|
| Three cards or more | 4 to 1 | 17.917%     |

# Mathematics

The tables above are generated from the game's code, which the tests below verify. Method and conventions: `docs/MATH.md`.

## Card source

The declared figures treat every card value from 1 to 10 as equally likely, as an infinite shoe deals them. The table deals from six decks, which moves the long-run figures by up to a tenth of a point per bet: see [The six-deck shoe](#).

## Landing on the target

The running total passes through a number  $n$  with chance  $h(n) = (h(n-1) + \dots + h(n-10)) / 10$ , with  $h(0) = 1$ : it gets there from one of the ten totals below, by a card of the right value. Up to 10 every earlier total is in reach, so  $h(n) = 1.1 \times h(n-1) = 1.1^{(n-1)} / 10$ , rising from 11.00% for a target of 2 to 23.58% for 10. From 11 the lowest totals drop out of reach and the chance falls back, to 15.94% for 11 and 16.53% for 12. Exact Hit pays most on 2 and least on 10, and 11 and 12 pay 5 to 1 because they are harder to hit than 9 or 10.

## Exact test

`moving-target.test.ts` computes every figure three ways, which must agree exactly, as fractions:

1. a memoised recursion over the running total and the number of cards, written from the rules alone; it also reproduces the published table (chance to hit each target, house edge per target and overall, the side bets' chances and edges);
2. the closed forms declared in the code;
3. the real game, run over the 36 rolls and every card sequence each can deal: 71,469 outcomes.

| Bet        | RTP, exactly                  | RTP     | House edge | Hit frequency | Volatility index $\sigma$ |
|------------|-------------------------------|---------|------------|---------------|---------------------------|
| Exact Hit  | 17307296056493/18000000000000 | 96.152% | 3.848%     | 17.301%       | 2.148                     |
| First Card | 11/12                         | 91.667% | 8.333%     | 9.167%        | 2.886                     |
| 3+ Cards   | 43/48                         | 89.583% | 10.417%    | 17.917%       | 1.917                     |

The volatility index is the standard deviation of the net result per unit staked; its square, the variance, is 1199/144 for First Card and 8471/2304 for 3+ Cards, and for Exact Hit a fraction of 28-digit terms worth 4.613.

## Monte Carlo test

`moving-target.math.test.ts` (run with `pnpm test:math`) plays 40,055,852 rounds of the full layout, all three bets at 0.50, dealt from an infinite shoe through the production game, with the seeded generator (seed `moving-target/infinite-shoe`). That count makes  $\pm 0.15$  percentage points equal to 3.29 standard errors for the most volatile bet, First Card. Every RTP must land within  $\pm 0.15$  pp of its declared value, and every hit frequency within 3.29 binomial standard errors.

| Bet        | Simulated RTP | Declared RTP | Difference | Standard error |
|------------|---------------|--------------|------------|----------------|
| Exact Hit  | 96.153%       | 96.152%      | +0.001 pp  | 0.034 pp       |
| First Card | 91.668%       | 91.667%      | +0.001 pp  | 0.046 pp       |
| 3+ Cards   | 89.578%       | 89.583%      | -0.006 pp  | 0.030 pp       |

## The six-deck shoe

A round deals several cards without replacement, so the real shoe does not return exactly the declared figures. Two tests measure by how much:

- `finite-shoe.test.ts` computes, exactly, the first round after a shuffle. Once a card is out, a second card of the same value is less likely (23 in 239 instead of 1 in 10), so totals reached through pairs get rarer. Even targets are reached through pairs more often, so they get harder.
- `six-deck.math.test.ts` plays 183,929,931 seeded rounds of the full layout on the real shoe, with its cut card and reshuffles (seed `moving-target/six-deck-shoe`), enough to measure each bet's shift to  $\pm 0.07$  pp. Over whole shoes the rounds interact: each round takes cards whose mix depends on the target it chased, the card that ends a round is more often a high one, and how many rounds a shoe deals depends on its cards. The test bounds every shift: 0.25 pp per bet, plus 3.29 standard errors per target.

| House edge | Infinite shoe (declared) | Six decks, first round (exact) | Six decks, long run | Shift (long run) |
|------------|--------------------------|--------------------------------|---------------------|------------------|
| Exact Hit  | 3.85%                    | 3.98%                          | 3.88%               | +0.04 ± 0.02 pp  |
| First Card | 8.33%                    | 8.33%                          | 8.39%               | +0.06 ± 0.02 pp  |
| 3+ Cards   | 10.42%                   | 10.62%                         | 10.30%              | −0.11 ± 0.01 pp  |

Exact Hit by target (house edge; the  $\pm$  is one standard error):

| Target | Infinite shoe | First round | Long run | Shift (long run) |
|--------|---------------|-------------|----------|------------------|
| 2      | 6.50%         | 6.82%       | 6.59%    | +0.09 ± 0.12 pp  |
| 3      | 3.20%         | 3.22%       | 2.93%    | −0.27 ± 0.08 pp  |
| 4      | 6.83%         | 7.11%       | 7.06%    | +0.23 ± 0.06 pp  |
| 5      | 4.83%         | 4.87%       | 4.70%    | −0.13 ± 0.05 pp  |
| 6      | 3.37%         | 3.62%       | 3.49%    | +0.12 ± 0.04 pp  |
| 7      | 2.56%         | 2.60%       | 2.49%    | −0.08 ± 0.04 pp  |
| 8      | 2.56%         | 2.78%       | 2.79%    | +0.22 ± 0.04 pp  |
| 9      | 3.54%         | 3.56%       | 3.50%    | −0.04 ± 0.04 pp  |
| 10     | 5.68%         | 5.84%       | 5.86%    | +0.18 ± 0.04 pp  |
| 11     | 4.38%         | 4.38%       | 4.27%    | −0.10 ± 0.07 pp  |
| 12     | 0.81%         | 1.06%       | 0.96%    | +0.15 ± 0.10 pp  |

In the long run the six-deck shoe makes the even targets a little harder and the odd ones a little easier; on Exact Hit as a whole the two nearly cancel (+0.04 pp). 3+ Cards gets slightly cheaper for the player (−0.11 pp) and First Card slightly dearer (+0.06 pp). No bet's house edge moves by more than a quarter of a point; a single target can move a little more (target 3, −0.27 pp).

Which figures a table returns depends on how it deals. An RNG table that draws every card independently (an infinite shoe) returns exactly the declared figures. One that reshuffles a virtual six-deck shoe before every round, much as a continuous shuffler does, returns the first-round column. A shoe dealt down to the cut card, as here and at a live table, returns the long-run column.

## Change log

- 2026-09-30: first playable version, with rules, declared math, exact and Monte Carlo tests, the six-deck shoe's shift and the table.
- 2026-09-30: renamed from its working title, with English names for the game and its bets.

# Mirror

*Your dice against the dealer's cards, hand for hand.* · original table game · **status: playable**

## Summary

The player places bets and rolls two dice: that is the player's hand. The dealer deals two cards from a shoe of aces to sixes: that is the dealer's hand. Both hands are read the same way, as a pair or as a sum, and compared. The main bet, **Mirror**, wins when the dice outrank the cards. Five side bets settle on the same two hands, one of them with a progressive meter. There are no decisions after the bets.

## Rules

### Equipment

- Two six-sided dice, rolled by the player.
- A shoe of six decks, each with the ace to the six in four suits (24 cards): 144 cards. The ace counts 1 and every other card its number; suits do not matter.
- The cut card sits with a quarter of the shoe left. When it comes out, the round finishes and the shoe is reshuffled before the next round. Every round deals two cards, so a shoe deals 54 rounds.

### A round

1. **Bet.** Place Mirror and, if you like, the side bets. Side bets need a Mirror bet. Limits: Mirror 0.50 to 250.00; each side bet 0.50 to 25.00.
2. **Roll.** Roll both dice: your hand.
3. **Deal.** The dealer deals two cards face down and turns them over: the dealer's hand.
4. **Settle.** Every bet settles on the two hands.

### Hands

A hand is two values from 1 to 6, and the dice and the cards are read the same way:

1. A **pair** beats any non-pair, and a higher pair beats a lower one. The table labels it **PAIR 4s**.
2. Between two non-pairs the higher **sum** wins; on equal sums, the higher single value wins. The table labels it **SUM 9 HIGH 6**.
3. Two hands that are still level hold the same two values: a tie.

From the strongest down: PAIR 6s to PAIR 1s, then SUM 11 HIGH 6 (6 and 5), SUM 10 HIGH 6, SUM 9 HIGH 6, SUM 9 HIGH 5, and so on down to SUM 3 HIGH 2 (2 and 1). Even PAIR 1s beats 6 and 5.

### Mirror

Mirror pays 1 to 1 when your dice outrank the dealer's cards. A tie loses.

### Side bets

| Bet            | Wins when                                                   | Pays                    |
|----------------|-------------------------------------------------------------|-------------------------|
| Tie            | the hands rank exactly equal: the same pair, or same values | 17 to 1                 |
| Equal Sums     | the dice and the cards add up to the same total             | 7 to 1                  |
| Pair vs Pair   | both hands are pairs                                        | 30 to 1                 |
| Perfect Mirror | both hands are the same pair                                | 200 to 1                |
| Double Sixes   | both hands are 6-6                                          | 1000 to 1 + meter share |

## The progressive meter

Double Sixes pays from a progressive meter on top of its fixed 1000 to 1.

- The meter starts at 5,000.00, its seed. 10% of every stake on Double Sixes goes into it as the bet is accepted, win or lose; the other 90% goes to the house. The main bet does not feed it.
- A hit pays 1000 to 1 on the stake plus the stake's share of the meter: stake ÷ 25.00 of it, so the whole meter at the 25.00 maximum and 2% of it at 0.50. The share is rounded down to the cent, and the round's own contribution is already in the meter.
- The meter keeps what the share leaves. If that is below the seed, the house tops it back up to 5,000.00.
- The demo keeps the meter in the browser between visits, and the lobby shows it on the table's card.

## Examples

- Dice 3 and 6 (SUM 9 HIGH 6), cards 4 and 5 (SUM 9 HIGH 5): the high 6 wins. Mirror pays 1 to 1, and Equal Sums 7 to 1.
- Dice 2 and 5, cards 5 and 2: a tie. Mirror loses; Tie pays 17 to 1 and Equal Sums 7 to 1.
- Dice 1 and 1 (PAIR 1s), cards 6 and 5 (SUM 11 HIGH 6): the pair wins. Mirror pays 1 to 1.
- Dice 4 and 4, cards 4 and 4: a tie, so Mirror loses; Tie, Equal Sums, Pair vs Pair and Perfect Mirror all win.
- 5.00 on Double Sixes with the meter at 6,000.00: the stake takes it to 6,000.50. Dice 6 and 6, cards 6 and 6: the bet pays 5,000.00 at 1000 to 1 plus a fifth of the meter, 1,200.10, and returns the stake. The meter keeps 4,800.40, which the house tops up to 5,000.00.

## Bets and payable

| Bet                   | RTP    | House edge | Hit frequency | Max exposure | Volatility index | Limits        |
|-----------------------|--------|------------|---------------|--------------|------------------|---------------|
| Mirror (main)         | 94.91% | 5.09%      | 47.45%        | 1×           | 0.999            | 0.50 – 250.00 |
| Tie (side)            | 91.67% | 8.33%      | 5.09%         | 17×          | 3.957            | 0.50 – 25.00  |
| Equal Sums (side)     | 90.12% | 9.88%      | 11.27%        | 7×           | 2.529            | 0.50 – 25.00  |
| Pair vs Pair (side)   | 86.11% | 13.89%     | 2.78%         | 30×          | 5.094            | 0.50 – 25.00  |
| Perfect Mirror (side) | 93.06% | 6.94%      | 0.46%         | 200×         | 13.645           | 0.50 – 25.00  |
| Double Sixes (side)   | 87.24% | 12.76%     | 0.08%         | —            | 33.348           | 0.50 – 25.00  |

RTP and house edge are per unit staked, pushes included. Hit frequency is the chance that a bet wins in a round. Max exposure is the largest net win per unit staked. The volatility index is the standard deviation of the net result per unit staked. A progressive bet's RTP excludes the seed of its meter, which the house funds, and its volatility index is taken with the meter at the seed; its terms follow its payable.

On the table's 6-deck shoe, every round returns exactly:

| Bet            | Hit frequency | RTP    | House edge | Edge vs declared |
|----------------|---------------|--------|------------|------------------|
| Mirror         | 47.74%        | 95.47% | 4.53%      | −0.57 pp         |
| Tie            | 5.11%         | 91.96% | 8.04%      | −0.29 pp         |
| Equal Sums     | 11.29%        | 90.29% | 9.71%      | −0.16 pp         |
| Pair vs Pair   | 2.68%         | 83.10% | 16.90%     | +3.01 pp         |
| Perfect Mirror | 0.45%         | 89.80% | 10.20%     | +3.25 pp         |
| Double Sixes   | 0.07%         | 84.54% | 15.46%     | +2.70 pp         |

## Mirror (main bet)

Wins when your dice outrank the dealer's two cards. A pair beats any non-pair and a higher pair a lower one; between non-pairs the higher sum wins, then the higher value. Ties lose.

| Outcome                         | Pays   | Probability |
|---------------------------------|--------|-------------|
| Dice outrank the dealer's cards | 1 to 1 | 47.454%     |

### Tie (side bet)

Wins when the two hands rank exactly equal: the same pair, or the same two values.

| Outcome              | Pays    | Probability |
|----------------------|---------|-------------|
| The hands rank equal | 17 to 1 | 5.093%      |

### Equal Sums (side bet)

Wins when the dice and the cards add up to the same total.

| Outcome  | Pays   | Probability |
|----------|--------|-------------|
| Same sum | 7 to 1 | 11.265%     |

### Pair vs Pair (side bet)

Wins when both hands are pairs.

| Outcome              | Pays    | Probability |
|----------------------|---------|-------------|
| Both hands are pairs | 30 to 1 | 2.778%      |

### Perfect Mirror (side bet)

Wins when both hands are the same pair.

| Outcome                      | Pays     | Probability |
|------------------------------|----------|-------------|
| Both hands are the same pair | 200 to 1 | 0.463%      |

### Double Sixes (side bet)

Wins when both hands are 6-6: 1000 to 1, plus a share of the progressive meter in proportion to the stake (the whole meter at the 25.00 maximum). 10% of every stake on this bet goes to the meter.

| Outcome            | Pays                                   | Probability |
|--------------------|----------------------------------------|-------------|
| Both hands are 6-6 | 1000 to 1 + stake ÷ 25.00 of the meter | 0.077%      |

| Meter                                     | Value                                                                                                                            |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Hit chance per round                      | 0.0772% (1 in 1,296); 6-deck shoe: 0.0745% (1 in 1,343)                                                                          |
| Fixed pays alone                          | 1000 to 1: RTP 77.24%, house edge 22.76%; 6-deck shoe: RTP 74.54%                                                                |
| Contribution to the meter                 | 10% of every stake                                                                                                               |
| RTP excluding the seed                    | 87.24% (house edge 12.76%): the fixed pays plus the contributions, which the meter pays out in the long run; 6-deck shoe: 84.54% |
| Meter share of a hit                      | stake ÷ 25.00 of the meter (the whole meter at 25.00)                                                                            |
| RTP with the meter at M                   | $77.24\% + M \div 32,400.00$ , for any stake                                                                                     |
| RTP with the meter at its seed (5,000.00) | 92.67%; 6-deck shoe: 89.43%                                                                                                      |
| Break-even meter                          | 7,375.00; 6-deck shoe: 8,548.91                                                                                                  |
| Average cycle                             | 1,296 rounds from hit to hit; 6-deck shoe: 1,343 rounds                                                                          |
| Meter at a hit, on average                | $5,000.00 + 129.6 \times \text{the mean stake}$ : 5,064.80 at 0.50, 5,129.60 at 1.00, 5,648.00 at 5.00, 8,240.00 at 25.00        |
| Seed cost to the house                    | 3.86 per round at 25.00 (15.43% of the stake): a cost, not part of the RTP                                                       |
| Max exposure per round                    | $1000 \times 25.00 + \text{the meter}$ : 30,000.00 with the meter at its seed, unbounded as it grows                             |

| Meter                        | Value  |
|------------------------------|--------|
| Volatility index at the seed | 33.348 |

The meter at a hit is exact on average when every cycle starts at the seed, as it does at 25.00, where each hit takes the whole meter; smaller stakes leave part of it behind, so their average is a little higher.

## Mathematics

The tables above are generated from the game's code, which the tests below verify. Method and conventions: `docs/MATH.md`.

### Card source

The declared figures treat each card as equally likely to be any value, as an infinite shoe deals them: 36 rolls against  $24 \times 24$  cards. The table deals from six decks, and there a pair of cards is rarer: once the first card is out, 23 of the 143 cards left share its value, not one in six. That moves every bet, as the six-deck table above shows: the pair bets lose about three points (Pair vs Pair returns 83.10%, Perfect Mirror 89.80%, Double Sixes' fixed pays 74.54%), and the main bet gains half a point (95.47%), because the dealer pairs less often.

Those six-deck figures are exact for every round, not only for the first after a shuffle. Mirror deals two cards a round and 54 rounds a shoe, whatever the cards, so the two cards of any round sit at fixed positions of a uniformly shuffled shoe: a uniform draw of two cards from the full 144. A table that reshuffles before every round would return the same figures.

### One published figure corrected

The brief published Equal Sums with  $P = 11.27\%$  and a house edge of  $9.85\%$ . The chance is right,  $146/1296$ ; at 7 to 1 the house edge is  $1 - 8 \times 146/1296 = 8/81 = 9.88\%$ . The tests assert  $9.88\%$ . Every other published figure is reproduced exactly.

### Exact tests

`mirror.test.ts` checks the declared figures three ways, which must agree exactly, as fractions:

1. it enumerates the  $36 \times 24 \times 24$  draws of the dice and two infinite-shoe cards with the ranking written from the rules, and reproduces the published table;
2. it checks that those are the fractions the bets declare (`bets.ts` computes them from the configuration, so retuning a payout keeps them right);
3. it runs the real game over all 20,736 draws: every RTP, hit frequency and volatility index equals its declared value.

For Double Sixes the game runs with the meter at zero for the fixed pays (exactly  $1001/1296$ ) and frozen at its seed for the volatility index ( $1201/1296$  per unit staked on average:  $1 + 1000 + 5,000.00 \div 25.00$  on a hit). The suite also checks that a hit pays every stake the same per unit ( $1000$  to 1 plus the meter  $\div 25.00$ , at 0.50, 1.00, 5.00 and 25.00), that the share is rounded down and the rest stays in the meter, and that the meter breaks even at 7,375.00.

`finite-shoe.test.ts` runs the game over every roll and every pair of cards a full six-deck shoe can deal (30,888 draws: the first card as a value, the second among the 143 left) and checks the six-deck figures. `game.test.ts` checks the meter's accounts over 200,000 rounds of mixed stakes: what it holds equals the seed plus the contributions and top-ups less the shares paid, to the millionth of a cent.

| Bet    | RTP, exactly | Hit frequency, exactly | Six decks, RTP exactly | Six decks, hit frequency |
|--------|--------------|------------------------|------------------------|--------------------------|
| Mirror | 205/216      | 205/432                | 4915/5148              | 4915/10296               |
| Tie    | 11/12        | 11/216                 | 263/286                | 263/5148                 |

| Bet                          | RTP, exactly | Hit frequency, exactly | Six decks, RTP exactly | Six decks, hit frequency |
|------------------------------|--------------|------------------------|------------------------|--------------------------|
| Equal Sums                   | 73/81        | 73/648                 | 1162/1287              | 581/5148                 |
| Pair vs Pair                 | 31/36        | 1/36                   | 713/858                | 23/858                   |
| Perfect Mirror               | 67/72        | 1/216                  | 1541/1716              | 23/5148                  |
| Double Sixes, fixed pays     | 1001/1296    | 1/1296                 | 161/216                | 23/30888                 |
| Double Sixes, excluding seed | 5653/6480    |                        | 161/216 + 1/10         |                          |

## The meter's economics

- **Where its money comes from.** Every stake on Double Sixes puts 10% in the meter; the house puts in the seed and every top-up.
- **Return excluding the seed.** The fixed pays return 77.24% and the meter pays back every contribution in the long run: 87.24% (84.54% on the six-deck shoe). This is the declared RTP.
- **Return at a given meter.** A hit pays meter ÷ 25.00 per unit staked, whatever the stake, so one round returns 77.24% + M ÷ 32,400.00 with the meter at M: 92.67% at the seed, 100% at 7,375.00 (8,548.91 on the six-deck shoe).
- **The meter at a hit.** A hit comes every 1,296 rounds on average (1,343 on the six-deck shoe). If every cycle starts at the seed, the meter at a hit averages the seed plus a cycle's contributions: 5,000.00 + 129.6 × the mean stake, 8,240.00 at a constant 25.00. With smaller stakes a hit takes only part of the meter and the rest carries over, so the meter settles higher: with 0.50, 1.00, 5.00 and 25.00 in turn (a mean of 7.875, for which the formula gives 6,020.60), it averaged 7,392.57 at a hit on the infinite shoe and 7,497.68 on the six-deck shoe.
- **The seed's cost to the house.** At a constant 25.00 every hit takes the whole meter, and the house re-seeds 5,000.00: 3.86 per round, 15.43% of the stake. That is more than the bet's 12.76% hold excluding the seed, so at a constant 25.00 the bet returns 102.67% with the seed counted. In the mixed-stake runs the top-ups came to 1.00 and 0.97 per round, and the bet returned 99.69% and 96.57% with them.

## Monte Carlo tests

Both runs play every bet in every round, with Double Sixes staked at 0.50, 1.00, 5.00 and 25.00 in turn and its meter live.

`mirror.math.test.ts` (run with `pnpm test:math`) plays 124,852,059 rounds on an infinite shoe, the source the declared figures assume, through the production game (seed `mirror/infinite-shoe`). That count makes ±0.15 percentage points 3.29 standard errors for Pair vs Pair, and the four bets up to its volatility must land within ±0.15 pp. Perfect Mirror and Double Sixes would need about 0.9 and 5.4 billion rounds for that; they are held to 3.29 of their own standard errors, and their exact tests are the proof. Every hit frequency must land within 3.29 binomial standard errors.

| Bet                          | Simulated RTP | Declared RTP | Difference | Standard error | Allowed   |
|------------------------------|---------------|--------------|------------|----------------|-----------|
| Mirror                       | 94.917%       | 94.907%      | +0.009 pp  | 0.009 pp       | ±0.150 pp |
| Tie                          | 91.642%       | 91.667%      | −0.025 pp  | 0.035 pp       | ±0.150 pp |
| Equal Sums                   | 90.136%       | 90.123%      | +0.012 pp  | 0.023 pp       | ±0.150 pp |
| Pair vs Pair                 | 86.085%       | 86.111%      | −0.026 pp  | 0.046 pp       | ±0.150 pp |
| Perfect Mirror               | 92.917%       | 93.056%      | −0.138 pp  | 0.122 pp       | ±0.402 pp |
| Double Sixes, fixed pays     | 76.984%       | 77.238%      | −0.253 pp  | 0.402 pp       | ±1.323 pp |
| Double Sixes, excluding seed | 86.984%       | 87.238%      | −0.253 pp  | 0.445 pp       | ±1.463 pp |

The RTP excluding the seed takes each round's payout less any top-up that round needed. The brief asked for it within  $\pm 0.3$  pp after 5,000,000 rounds on the real shoe. The real shoe returns 84.54%, not 87.24%, so that run checks the six-deck figure; and with a volatility index near 33, a  $\pm 0.3$  pp check at 3.29 standard errors would take over a billion rounds. The exact tests pin both parts instead: the fixed pays exactly, and the contributions through the meter's accounts.

`six-deck.math.test.ts` plays 31,213,015 rounds on the real shoe with its cut card and reshuffles (seed `mirror/six-deck-shoe`), enough to hold Pair vs Pair to  $\pm 0.3$  pp, and every bet to 3.29 of its standard errors around its exact six-deck figure.

| Bet                          | Declared | Six decks, exact | Simulated | Difference | Standard error |
|------------------------------|----------|------------------|-----------|------------|----------------|
| Mirror                       | 94.907%  | 95.474%          | 95.485%   | +0.011 pp  | 0.018 pp       |
| Tie                          | 91.667%  | 91.958%          | 91.819%   | −0.139 pp  | 0.071 pp       |
| Equal Sums                   | 90.123%  | 90.287%          | 90.192%   | −0.095 pp  | 0.045 pp       |
| Pair vs Pair                 | 86.111%  | 83.100%          | 82.967%   | −0.133 pp  | 0.090 pp       |
| Perfect Mirror               | 93.056%  | 89.802%          | 89.287%   | −0.515 pp  | 0.239 pp       |
| Double Sixes, fixed pays     | 77.238%  | 74.537%          | 74.309%   | −0.228 pp  | 0.791 pp       |
| Double Sixes, excluding seed | 87.238%  | 84.537%          | 84.307%   | −0.230 pp  | 0.878 pp       |

## Card counting

A player who tracks the cards dealt knows the shoe's composition before every round. The six-deck run computes each bet's exact expectation for that composition, round by round:

| Bet                          | Rounds favouring the counter | Counter's edge in them | Break-even bet spread |
|------------------------------|------------------------------|------------------------|-----------------------|
| Mirror                       | 10.80%                       | 2.8%                   | 1 to 16               |
| Tie                          | none                         | —                      | —                     |
| Equal Sums                   | under 1 in 10,000            | 0.7%                   | —                     |
| Pair vs Pair                 | 0.06%                        | 3.4%                   | 1 to 8,458            |
| Perfect Mirror               | 0.61%                        | 3.5%                   | 1 to 486              |
| Double Sixes, fixed pays     | 11.77%                       | 22.8%                  | 1 to 10.5             |
| Double Sixes, with the meter | 41.11%                       | 26.3%                  | 1 to 1.3              |

The break-even spread is how much more a perfect counter must stake in the favourable rounds than in the others (one unit) to break even. The main bet is exposed: some compositions leave the dealer's hand weak enough that the dice are more likely than not to win. Double Sixes is the most exposed, since its chance follows the sixes left; with the meter as it stood in this run (above break-even much of the time), a counter needs almost no spread. A table that reshuffles before every round, as a continuous shuffler does, removes the exposure and changes no figure on this sheet.

## Change log

- 2026-09-30: first playable version, with rules, declared math, exact and Monte Carlo tests on an infinite and on the six-deck shoe, the progressive meter's economics, card counting exposure and the table.
- 2026-09-30: renamed from its working title, with English names for the game and its bets; the progressive bet is Double Sixes.

# Lock & Roll

*Lock a die, roll the other, beat the cards.* · original table game · **status: playable**

## Summary

---

The player places one bet and rolls two dice. Then comes the one decision in the demo: **Stand** on the roll, or **Lock**: keep one die and roll the other once more, for the Lock fee of 40% of the bet, free on 1-1. The dealer then deals two cards from a shoe of aces to sixes, and the bet wins even money when the dice add up to more than the cards. A tie goes to the house.

## Rules

---

### Equipment

- Two six-sided dice, rolled by the player.
- A shoe of six decks, each with the ace to the six in four suits (24 cards): 144 cards. The ace counts 1 and every other card its number; suits do not matter.
- The cut card sits with a quarter of the shoe left. When it comes out, the round finishes and the shoe is reshuffled before the next round. Every round deals two cards, so a shoe deals 54 rounds.

### A round

1. **Bet.** Place the Lock & Roll bet: 0.50 to 250.00. There are no side bets.
2. **Roll.** Roll both dice.
3. **Decide, once.**
  - **Stand:** keep the roll.
  - **Lock:** lock one die and roll the other once more. The Lock fee is 40% of the bet (0.40 on a 1.00 bet), taken at once and never returned, whatever the result. A roll of 1-1 re-rolls for free. The re-rolled die is final.
4. **Deal.** The dealer deals two cards face up.
5. **Settle.** If the dice add up to more than the two cards, the bet wins 1 to 1. An equal or lower total loses. The Lock fee is not part of the bet: nothing is paid on it.

The Lock fee is rounded up to the cent, which only matters for bets that are not multiples of 0.05; the table's chips (0.50 and up) always make it exact. If the table closes before you decide (you leave the page, say), the round finishes as if you had chosen Stand: no Lock fee is taken without your choice.

**Not offered in this version:** *Lock & Double*, locking a die, re-rolling and doubling the bet. The engine keeps a place for it.

### Examples

- Bet 1.00, roll 6-1. Lock the 6 and re-roll the 1 for 0.40: it lands on 4, for 10. The dealer deals a 5 and a 3, for 8. The bet wins 1.00; with the Lock fee spent, the round nets +0.60.
- Bet 1.00, roll 1-1. The re-roll is free: lock a 1, and the other lands on 5, for 6. The dealer deals two 3s, for 6. A tie goes to the house: -1.00.
- Bet 1.00, roll 6-2. Stand on 8, which beats the dealer 7 times in 12. The dealer deals a 6 and a 5: -1.00. Standing was still right: a re-roll keeping the 6 is worth less, once the Lock fee is paid.

## Strategy hint

The table can show the reference strategy's choice for your roll. The **Strategy hint** switch beside the decision is off by default. It is a demonstration feature for evaluators, and it shows exactly what the table's autoplay does.

## Bets and payable

| Bet                | RTP    | House edge | Hit frequency | Max exposure | Volatility index | Limits        |
|--------------------|--------|------------|---------------|--------------|------------------|---------------|
| Lock & Roll (main) | 95.93% | 4.07%      | 56.85%        | 1×           | 1.034            | 0.50 – 250.00 |

RTP and house edge are per unit staked, pushes included. Hit frequency is the chance that a bet wins in a round. Max exposure is the largest net win per unit staked. The volatility index is the standard deviation of the net result per unit staked. The figures assume the strategy below, and count any fee paid for a choice against the return: a fee is never returned, and it is not a stake.

On the table's 6-deck shoe, every round returns exactly:

| Bet         | Hit frequency | RTP    | House edge | Edge vs declared |
|-------------|---------------|--------|------------|------------------|
| Lock & Roll | 56.87%        | 95.97% | 4.03%      | −0.04 pp         |

## Lock & Roll (main bet)

Wins 1 to 1 when your dice add up to more than the dealer's two cards; a tie loses. After the roll you may lock one die and roll the other once more, for the Lock fee of 40% of the bet, taken at once and never returned (a roll of 1-1 re-rolls for free).

| Outcome                      | Pays   | Probability |
|------------------------------|--------|-------------|
| Dice beat the dealer's cards | 1 to 1 | 56.854%     |

## Strategy

After the roll: Stand, or Lock (keep one die and roll the other once more) for the Lock fee, 40% of the bet, taken at once and never returned; a roll of 1-1 re-rolls for free. A re-roll is always worth more with the higher die locked. The reference strategy re-rolls on 1-1, 2-1, 3-1, 4-1, 5-1, 6-1, 3-2, 4-2, 5-2 and stands on the other 12 rolls.

| Roll | Play                               | Stand  | Lock   | Chance |
|------|------------------------------------|--------|--------|--------|
| 1-1  | Lock a 1, re-roll the other (free) | −1.000 | −0.676 | 2.78%  |
| 2-1  | Lock the 2, re-roll the 1          | −0.944 | −0.881 | 5.56%  |
| 3-1  | Lock the 3, re-roll the 1          | −0.833 | −0.650 | 5.56%  |
| 4-1  | Lock the 4, re-roll the 1          | −0.667 | −0.400 | 5.56%  |
| 5-1  | Lock the 5, re-roll the 1          | −0.444 | −0.150 | 5.56%  |
| 6-1  | Lock the 6, re-roll the 1          | −0.167 | +0.081 | 5.56%  |
| 2-2  | Stand                              | −0.833 | −0.881 | 2.78%  |
| 3-2  | Lock the 3, re-roll the 2          | −0.667 | −0.650 | 5.56%  |
| 4-2  | Lock the 4, re-roll the 2          | −0.444 | −0.400 | 5.56%  |
| 5-2  | Lock the 5, re-roll the 2          | −0.167 | −0.150 | 5.56%  |
| 6-2  | Stand                              | +0.167 | +0.081 | 5.56%  |
| 3-3  | Stand                              | −0.444 | −0.650 | 2.78%  |
| 4-3  | Stand                              | −0.167 | −0.400 | 5.56%  |
| 5-3  | Stand                              | +0.167 | −0.150 | 5.56%  |
| 6-3  | Stand                              | +0.444 | +0.081 | 5.56%  |
| 4-4  | Stand                              | +0.167 | −0.400 | 2.78%  |

| Roll | Play  | Stand         | Lock   | Chance |
|------|-------|---------------|--------|--------|
| 5-4  | Stand | <b>+0.444</b> | −0.150 | 5.56%  |
| 6-4  | Stand | <b>+0.667</b> | +0.081 | 5.56%  |
| 5-5  | Stand | <b>+0.667</b> | −0.150 | 2.78%  |
| 6-5  | Stand | <b>+0.833</b> | +0.081 | 5.56%  |
| 6-6  | Stand | <b>+0.944</b> | +0.081 | 2.78%  |

Each value is the expected net result per unit of the main bet, the fee included (Lock keeps the higher die). The best choice is in bold: it is the strategy the declared figures assume.

## What the strategy returns

| Figure                    | These rules       | Without the free 1-1 | These rules, 6-deck shoe |
|---------------------------|-------------------|----------------------|--------------------------|
| RTP                       | 95.93%            | 95.03%               | 95.97%                   |
| House edge                | 4.07%             | 4.97%                | 4.03%                    |
| Element of risk           | 3.45%             | 4.22%                | 3.42%                    |
| Win frequency             | 56.85%            | 56.40%               | 56.87%                   |
| Lock, share of rounds     | 47.22%            | 44.44%               | 47.22%                   |
| Fee paid, share of rounds | 44.44%            | 44.44%               | 44.44%                   |
| Average fee per round     | 17.78% of the bet | 17.78% of the bet    | 17.78% of the bet        |
| Volatility index          | 1.034             | 1.034                | 1.034                    |

The house edge is the loss per unit of the main bet, fees included. The element of risk divides the same loss by everything the player pays: the bet and the fees.

## Mathematics

The tables above are generated from the game's code, which the tests below verify. Method and conventions: `docs/MATH.md`.

### The Lock fee in the figures

The Lock fee is not a stake: nothing pays on it, and it is never returned. The figures count it against the return, per unit of the bet:  $\text{RTP} = (\text{payouts} - \text{fees}) \div \text{bets}$ . That gives the house edge of 4.07%. The same loss divided by everything the player pays, the bet and the fees, is the element of risk, 3.45%. Counted that way, as if the fees were money wagered, the game returns 96.55%. The RTP monitor at the table counts the fees against the return as well, so it converges on 95.93%.

### The strategy

After the roll the player knows their total and compares two numbers: the value of standing on it ( $2 \times$  the chance that the dealer's two cards total less, minus 1) and the value of a re-roll (the same for each face of the re-rolled die, on average, minus the fee). The strategy card above lists both for each of the 21 distinct rolls. The best play follows:

- A re-roll always keeps the higher die: the higher the die kept, the better the re-roll.
- It pays to re-roll when the lower die is a 1 or a 2 and the other is not too high: 2-1 to 6-1, 3-2, 4-2 and 5-2.
- Never re-roll a double, except 1-1 when it is free: 2-2 stands on 4 (−0.833) rather than pay for a re-roll worth −0.881.
- 6-2 stands: 8 wins 7 times in 12 (+0.167), a re-roll keeping the 6 is worth +0.081 after the fee.
- The closest calls are 3-2 and 5-2: a re-roll is better by 1/60 of the bet.

The free 1-1 is worth 35/3888 of the bet to the player (0.90 points of house edge), and without it, 1-1 stands: a sure loss beats paying 0.40 for a re-roll worth  $-0.676$ . The strategy re-rolls on 17 rolls in 36 (47.2%) with the free 1-1 and on 16 in 36 (44.4%) without it, and pays the fee on 16 rolls in 36 either way.

## Card source

The declared figures treat each card as equally likely to be any value, as an infinite shoe deals them. The table deals from six decks, where a pair of cards is a little rarer: once the first card is out, 23 of the 143 cards left share its value, not one in six. The dealer's totals move toward the middle, which helps the dice a little: the house edge is 4.03% on the six-deck shoe, against 4.07% declared. The strategy is the same on both shoes.

Those six-deck figures are exact for every round, not only for the first after a shuffle. Lock & Roll deals two cards a round and 54 rounds a shoe, whatever the dice and the decisions, so the two cards of any round sit at fixed positions of a uniformly shuffled shoe: a uniform draw of two cards from the full 144.

## Exact tests

- `strategy.test.ts` derives the strategy from scratch. It values standing and both re-rolls on each of the 21 rolls by enumerating the dealer's two cards and the faces of the re-rolled die, with exact fractions, and picks the best. That is the declared strategy on every roll, and the engine's `strategy.ts` computes the same choices. The house edge follows exactly: 791/19440 with the free 1-1 and 161/3240 without it.
- `lock-and-roll.test.ts` runs the real game with the reference strategy over every roll, every re-rolled face and every pair of infinite-shoe cards (69,696 draws). The RTP, the fees, the win frequency and the volatility equal the declared figures exactly. The rules without the free 1-1 return 3079/3240, and every other strategy returns less: always standing returns 575/648 (88.73%).
- `finite-shoe.test.ts` runs the game over every roll, re-roll and pair of cards a full six-deck shoe can deal (103,818 draws) and checks the six-deck figures.

| Figure                   | These rules, exactly | Without the free 1-1 | Six decks, exactly |
|--------------------------|----------------------|----------------------|--------------------|
| RTP                      | 18649/19440          | 3079/3240            | 148219/154440      |
| House edge               | 791/19440            | 161/3240             | 6221/154440        |
| Win frequency            | 4421/7776            | 731/1296             | 35135/61776        |
| Re-rolls                 | 17/36                | 4/9                  | 17/36              |
| Fee paid                 | 4/9                  | 4/9                  | 4/9                |
| Average fee (of the bet) | 8/45                 | 8/45                 | 8/45               |
| Element of risk          | 791/22896            | 161/3816             | 6221/181896        |

## Monte Carlo tests

Both runs play the production game with the reference strategy deciding every round, the same bot the table's autoplay uses. `lock-and-roll.math.test.ts` (run with `pnpm test:math`) plays 5,144,842 rounds on an infinite shoe (seed `lock-and-roll/infinite-shoe`): enough to make  $\pm 0.15$  percentage points 3.29 standard errors at a volatility of 1.034. The frequencies must land within 3.29 binomial standard errors.

| Figure            | Declared | Simulated | Difference  | Allowed        |
|-------------------|----------|-----------|-------------|----------------|
| RTP               | 95.931%  | 95.874%   | $-0.057$ pp | $\pm 0.150$ pp |
| Win frequency     | 56.854%  | 56.835%   | $-0.019$ pp | $\pm 0.072$ pp |
| Lock (re-rolls)   | 47.222%  | 47.270%   | $+0.048$ pp | $\pm 0.072$ pp |
| Free 1-1 re-rolls | 2.778%   | 2.779%    | $+0.002$ pp | $\pm 0.024$ pp |
| Fee paid          | 44.444%  | 44.491%   | $+0.047$ pp | $\pm 0.072$ pp |

six-deck.math.test.ts plays as many rounds on the real shoe, with its cut card and reshuffles (seed lock-and-roll/six-deck-shoe , 95,275 shuffles). It returned 95.861% against the exact six-deck 95.972%: −0.111 pp, 2.4 standard errors of 0.046 pp, inside both bounds the test sets ( $\pm 0.15$  pp and 3.29 standard errors).

### Card counting

The dealer's cards are dealt face up, so a player who tracks them knows the shoe's composition before every round. The six-deck run computes the exact expectation for that composition, round by round:

| Player                  | Rounds favouring the counter | Counter's edge in them | Break-even bet spread |
|-------------------------|------------------------------|------------------------|-----------------------|
| The reference strategy  | 21.07%                       | 4.6%                   | 1 to 5.1              |
| Deciding with the count | 21.09%                       | 4.7%                   | 1 to 5.0              |

The break-even spread is how much more a perfect counter must stake in the favourable rounds than in the others (one unit) to break even. Lock & Roll is exposed: the dealer's total follows the cards left, and a shoe short of high cards makes the dice favourites in about one round in five. Adjusting the decisions to the count adds almost nothing. A table that reshuffles before every round, as a continuous shuffler does, removes the exposure; its figures are the six-deck figures above.

### Change log

- 2026-09-30: first playable version, with rules, the decision and its fee, the strategy computed by expected value, exact and Monte Carlo tests on an infinite and on the six-deck shoe, card counting exposure and the table.
- 2026-09-30: renamed from its working title, with English names for the game and its bet; the choices are Stand and Lock, and the re-roll costs the Lock fee.